numerous exchanges and trading venues, we find the same patterns; costs are related to trading characteristics. Different exchanges provide different needs for investors. We did not find that any one structure was less efficient or more costly than another. One factor, however, that stuck out as having a very large correlation with cost is liquidity. Both historically and in today's markets, we have found that those venues with greater liquidity were associated with lower trading costs.

PRICING MODELS

Currently there are three different pricing models used by trading venues: maker-taker, taker-maker, also known as inverted pricing models, and commission based. All of the different models are put forth to attract as much liquidity as possible to their exchange.

Maker-Taker Model. In the maker-taker model, the maker is the investor posting and providing liquidity. In return, the maker is paid a rebate to provide liquidity to the exchange, and the taker of liquidity is charged a larger transaction fee than the rebate paid to the maker. The rebate is only provided if a transaction takes place. Thus, investors are incentivized to provide liquidity to the particular exchange or venue.

Taker-Maker (Inverted) Model. In the taker-maker model, the investor posting the order is charged a fee for supplying liquidity and the investor taking the liquidity is provided a rebate. Why would investors pay to provide liquidity when they could enter liquidity on another venue and receive a rebate for this service? The answer is simple. Suppose the investor is at the end of a long price-time priority queue. The investor could increment their bid or offer price costing them a full price increment. However, a better option that allows them to jump to the top of the queue is to place the order on a taker-maker exchange, where the rebate charged would still be less than the price increment. Hence, liquidity taking investors have two options; one being charged a fee to take liquidity and another being paid to take liquidity. At the same price, rational investors would always select the option where they would be paid a rebate as opposed to having to pay a rebate. The taker-maker model allows investors to jump to the front of the line for a small fee. As long as this fee is less than the full price increment (less any expected rebate the investor is expected to receive from the transaction) they would select the pay to post option. The option proves valuable in situations where investors are looking to improve the best market price or utilize a market order where they would pay the entire spread and rebate in addition to crossing the spread.